MIL-B-60298A(MU)
18 June 1970
SUPERSEDING
MIL-B-60298(MU)
29 September 1966

MILITARY SPECIFICATION

BATTERY, NICKEL-CADMIUM, VENTED: 10541198

1. SCOPE

1.1 This specification covers a portable, vented, nickel-cadmium battery which is the primary power source for the Range Finder, Laser, AN/GVS-3 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Fefinitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10541198	Battery, Nickel-Cadmium, Vented
C10541118	Plug

PACKAGING DATA SHEET

10541198	Packaging of Battery, Nickel-Cadmium, Vented

FSC 1240

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The battery shall be manufactured in accordance with Drawing F10541198 and drawings pertaining thereto.

3.3 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Part identification and marking
d. Workmanship
e. Soldering electrical connections

3.4 Environmental.

3.4.1 Vibration.- The battery shall withstand vibration for 15 minutes each in 3 mutually perpendicular directions. Vibration in each direction shall be at frequencies varied logarithmically from 5 to 500 cycles per second (cps). The double amplitude of applied vibration shall be 0.036 inch at lower frequencies and shall be maintained at, but not exceed, a 5g level at higher frequencies. Subsequent to vibration, the battery shall show no evidence of physical damage, there shall be no leakage of electrolyte, and the battery shall meet the requirements of 3.5 through 3.8.

3.4.2 Shock.- The battery shall withstand a total of 18 shock impulses, 3 in each direction of 3 mutually perpendicular axes. Each shock impulse shall be a half sine wave with a time duration of 11 ± 1 milliseconds. The peak amplitude of each shock impulse shall be 75g's. Subsequent to shock, the battery shall show no evidence of physical damage, there shall be no leakage of electrolyte and the battery shall meet the requirements of 3.5 through 3.8.

3.4.3 Storage temperatures.- The battery shall withstand thermal stabilization at temperatures of -80°Fahrenheit (F) and +160°F. Subsequent to exposure to the temperatures specified herein, the battery shall show no evidence of physical damage, there shall be no leakage of electrolyte, and the battery shall meet the requirements of 3.5 through 3.8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

3.4.4 Altitude.

3.4.4.1 Operating.- The battery shall meet the requirements of 3.5.2 when subjected to the following conditions in an uninterrupted sequence:

a. Elevated from ground level to the conditions encountered at an altitude of 10,000 feet above sea level within 2 minutes
b. Maintained at the 10,000 foot level for a minimum of 6 minutes
c. Returned to ground level within 2 minutes

During the cycle"a" through "c", there shall be no fluctuation of voltage and current. This requirement shall be met with the battery held in an upright position.

3.4.4.2 Non-operating.- The battery when in a discharged condition, shall withstand storage at an altitude of 50,000 feet above sea level for a duration of not less than 1 hour, during which or subsequent to storage, the battery shall show no evidence of bulging of the case or leakage of electrolyte with the battery held in any position.

3.5 Performance.

3.5.1 Full capacity charge.- From an initial open circuit voltage of 7.0 volts, each section of the battery shall be capable of attaining the voltage specified below when recharged at a constant current of 500 ma to 120% of the 1.5 ampere hour capacity of sections 10541201 and 10558189 and the 2.0 ampere hour capacity of section 10541200.

11.9 ± 0.7V	at	70°F
11.2 ± 0.7V	at	155°F
12.3 ± 0.7V	at	-25°F

3.5.2 Full capacity discharge.- The requirements of 3.5.2.1 and 3.5.2.2 shall be met when the battery is discharged at the same temperature it was charged in 3.5.1. The discharge of the battery section shall begin from an initial open circuit voltage of not less than 8.75V.

3.5.2.1 Room and high temperature discharge.- When thermally stabilized at temperatures of 155°F and 70°F, the battery shall provide the capacities at the rates specified below:

a. A constant current of 1.0 ampere and a minimum voltage of 6.65 volts shall be provided to a resistive load between pins A and B for not less than 60 minutes.

b. A constant current of 1.0 ampere and a minimum voltage of 6.65 volts shall be provided to a resistive load between pins C and D for not less than 60 minutes.

c. A constant current of 1.3 amperes and a minimum voltage of 6.65 volts shall be provided to a resistive load between pins E and F for not less than 60 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

3.5.2.2 Cold temperature discharge.- When thermally stabilized at -25°F, the battery shall provide the capacities at the rates specified below:

a. A constant current of 1.0 ampere and a minimum voltage of 6.30 volts shall be provided to a resistive load between pins A and B for not less than 10 minutes.

b. A constant current of 1.0 ampere and a minimum voltage of 6.30 volts shall be provided to a resistive load between pins C and D for not less than 10 minutes.

c. A constant current of 1.30 amperes and a minimum voltage of 6.30 volts shall be provided to a resistive load between pins E and F for not less than 10 minutes.

3.5.3 Retention of charge.- The battery shall retain 60% of its full ampere hour capacity after storage at temperatures between 70° and 80°F for a period of 30 days after being charged to its full capacity as in 3.5.1.

3.5.4 Operational cycling.- The ampere-hour capacity degradation shall be less than 25 percent after being subjected to a minimum of 100 cycles of charging and discharging (one charge and one discharge constitutes one cycle). The 100 cycles shall consist of the following:

a. Five cycles in which the battery shall be discharged to the specified minimum voltage (7V per section) and charged to full capacity while stabilized at -25°F.

b. Five cycles in which the battery shall be discharged to the specified minimum voltage (7V per section) and charged to full capacity while stabilized at +155°F.

c. Five cycles in which the battery shall be discharged to the specified minimum voltage (7V per section) and charged to full capacity while at a temperature between 70° and 80°F.

d. Eighty-five (85) cycles in which the battery shall be discharged for a period of 15 minutes after which the battery shall be charged to full capacity at a temperature between 70° and 80°F.

The discharge portion of each cycle shall be at the applicable rate specified in 3.5.2 and the charge portion of each cycle at the current rate specified in 3.5.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

3.5.5 Temperature sensing resistor.- The resistance between pins G and H shall be as specified below at the temperatures indicated:

Resistance	Temperature
800 ohms ± 10%	+70°F
670 ohms ± 10%	+155°F
1300 ohms ± 10%	-25°F

3.5.6 Battery vent.- The battery vent shall equalize the increases in internal battery pressure encountered during charging with the atmospheric pressure.

3.6 Continuity.- Continuity shall be in accordance with the wiring diagram on Drawing F10541198.

3.7 Insulation resistance.- Unless otherwise specified on applicable drawing, all insulation between electrically isolated circuits, any terminals not electrically connected and between ungrounded terminals and the battery case shall have a minimum resistance of 1000 megohms with 500 volts direct current applied.

3.8 Interface requirements.- The battery connector, 10541202, shall interface with Plug, 10541118.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 assemblies manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS
(100% INSPECTION)

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Full capacity charge	3.5.1	4.6.6
102.	Full capacity discharge (70°F)	3.5.2.1	4.6.7
103.	Temperature sensing resistor	3.5.5	4.6.10
104.	Battery vent	3.5.6	4.6.11
105.	Continuity	3.6	4.6.12
106.	Insulation resistance	3.7	4.6.13
107.	Interface	3.8	4.6.14

NOTE: The tests in Table I shall be conducted at a temperature between +70°F and +80°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Three batteries shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable dwgs & specs - visual
302.	Fabrication	3.2	Applicable dwgs & specs - visual
303.	General specification	3.3	MIL-F-13926 - visual

NOTE: The tests in Table II shall be conducted at a temperature between +70°F and +80°F.

4.4.2 Environmental.- One battery shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Vibration	3.4.1	4.6.3
305.	Shock	3.4.2	4.6.2
306.	Storage temperature	3.4.3	4.6.1.1
307.	Altitude, operating	3.4.4.1	4.6.4
308.	Altitude, non-operating	3.4.4.2	4.6.5
309.	Full capacity charge (+155°F)	3.5.1	4.6.6
310.	Full capacity charge (-25°F)	3.5.1	4.6.6
311.	Full capacity discharge (+155°F)	3.5.2.1	4.6.7
312.	Full capacity discharge (-25°F)	3.5.2.2	4.6.7
313.	Retention of charge (+70°F)	3.5.3	4.6.8
314.	Operational cycling	3.5.4	4.6.9

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted items have been approved (see 4.3.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.2, the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6.

NOMENCLATURE	DESCRIPTION
1. Voltmeter D.C.	Minimum sensitivity of 20,000 ohms per volt capable of measuring to 15 VDC with 1% accuracy.
2. Ammeter, D.C.	Accuracy ±1%, capable of measuring the discharge currents (see 3.5.2).
3. Power supply (constant current)	Capable of supplying 500ma at voltages between 0 and 15 VDC.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

4.6 Test methods and procedures.

4.6.1 Environmental.- The battery shall be thoroughly examined for physical defects after each environmental test. A visual and tactile examination shall be made for missing parts, leakage of electrolyte and physical damage. Upon completion of all environmental tests, the battery shall be subjected to the tests specified in 4.6.6 through 4.6.14.

4.6.1.1 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperature specified in 3.4.3.

4.6.2 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 except the shock pulse shall comply with the wave form, time duration and peak "G" value specified in 3.4.2.

4.6.3 Vibration.- The test equipment utilized in this test shall conform to the vibration apparatus requirements specified in MIL-STD-810. The battery shall be firmly mounted to the vibration table. The vibration equipment shall provide the conditions of acceleration amplitude, frequency and duration specified in 3.4.1. The vibration test shall then be performed in accordance with Method 514 Procedure I, Part 1, of MIL-STD-810.

4.6.4 Altitude, operating.- Perform this test in accordance with Procedure I of the altiutde test in MIL-F-13926, with conditions specified in 3.4.4.1. The discharge of the battery during the altitude cycle shall be monitored with the ammeter and voltmeter specified in 4.5.3 placed external to the altitude chamber. The battery shall conform to the requirements specified in 3.4.4.1.

4.6.5 Altitude, non-operating.- Conduct this test in sequence to the test in 4.6.4, after the battery has been discharged to an open circuit voltage of 7.0V. Perform this test in accordance with Procedure II of the altitude test in MIL-F-13926, with conditions specified in 3.4.4.2. The battery shall conform to the requirements of 3.4.4.2 when rotated to any position.

4.6.6 Full capacity charge.- Discharge the 3 sections of the battery at the rates specified in 3.5.2.1 and 3.5.2.2 for a period of time sufficient to lower the terminal voltage of each section to 7.0 ± 0.1V. Immediately charge section 10541201 through pin A (+) and B (-), and section 10558189 through pin C (+) and D (-) at 500 ma for 3.6 hours and charge section 10541200 through pin E (+) and pin F (-) at 500 ma for 4.8 hours using a constant current source (Item 3 of 4.5.3). At the end of this charging time, the terminal voltage measured with the voltmeter (Items 1 of 4.5.3) shall conform to 3.5.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

4.6.7 Full capacity discharge.- Connect battery sections which meet the requirements of 3.5.2 to a load which will maintain the current drain on the battery as specified in 3.5.2.1 and 3.5.2.2. Discharge the battery for the specified period of time at the specified temperature. The terminal voltage after the battery has been discharged as above shall be not less than those values specified in 3.5.2.1 and 3.5.2.2 while supplying the specified load currents.

4.6.8 Retention of charge.- Charge the battery at 70°F as specified in 3.**5.1.** Store the battery at the temperature specified in 3.5.3 for the 30 day period. Discharge the battery as in 4.6.7 for 36 minutes. After discharge, the battery terminal voltage under load shall not be less than the values specified in 3.5.2.1.

4.6.9 Operational cycle.- Charge and discharge the battery in accordance with 3.5.1 and 3.5.2 as specified in 3.5.4. Following this, charge the battery at 70°F according to 4.4.6. Discharge the battery as specified in 4.6.7 for 45 minutes at 70°F. The terminal voltage after the battery has been discharged as above shall conform to 3.5.2.1.

4.6.10 Temperature sensing resistor.- Use test equipment in accordance with the "Test Facilities" requirements of MIL-F-13926. Thermally stabilize the battery at each specified temperature for a period of 2 hours. At the end of each 2 hour period and while still at the specified temperature, measure the resistance of the temperature sensing resistor with an ohmmeter. The battery shall conform to the requirements of 3.5.5 at the temperature indicated.

4.6.11 Battery vent.- The battery vent shall be subjected to a visual examination during all temperature and altitude tests for evidence of malfunction. The battery shall conform to the requirements of 3.5.6.

4.6.12 Continuity.- Standard measuring equipment shall be used to determine electrical continuity requirements.

4.6.13 Insulation resistance.- Measure the insulation resistance between mutually insulated parts by means of a megohm bridge or other approved method.

4.6.14 Interface requirements.- All pins of the Battery Connector C10541202 shall engage the Plug D10541118 and continuity with all connector pins shall be indicated when fully engaged.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10541198. The level of protection shall be as specified in the procurement document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use.- The battery is the primary power source for the Range Finder, Laser, AN/GVS-3.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Applicable packaging data sheet
c. Level of protection required
d. Provisions for First Article testing

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A163

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:26Z
Check the source to verify that this is the current version before use.